

600-User Load Test Report

Target: LG Magazine (lg_magazine) · Environment: LG production (mybook.lgacademy.com / 203.247.146.226) · Next.js 16 · Docker 3 replicas · nginx · 1 Gbps · Date: 2026-06-08 · Tool: k6

The production server handles 600 concurrent users with virtually zero errors.

A fresh test ran 600 users against live production (2-min gradual ramp, then held at 600). Result: **99.94% success**, warm-navigation p95 of **263 ms**, using only ~10% of the 1 Gbps line. The Thursday class (600 people) is comfortably within capacity.

99.94% success rate (0.06% failed)	263 ms p95, warm navigation	104.1 Mbps peak bandwidth (~10% of 1 Gbps)	27,287 complete user sessions
--	---	--	---

1. Test setup

Parameter	Value
Concurrent users (VUs)	600
Ramp-up	2 min (0 → 600, mimics a class arriving)
Hold at peak	2 min at 600 concurrent
User behavior	Cold first visit (full page + assets) → warm navigation between chapters, with 2-5s think time
Scope	Page loads, static assets, auth check — AI generation endpoint NOT exercised (no token usage)

2. Results

Metric	Result	Meaning
Success rate	99.94%	0.06% failed — negligible
Total requests	90,862	350.2/s sustained
Complete sessions	27,287	full user journeys finished
p95 — warm navigation	263 ms	chapter-to-chapter, cached
p95 — overall	445 ms	avg 301 ms
p95 — home / navigate / auth	132 / 133 / 131 ms	core responses fast
p95 — cold first-visit assets	20 s	first load pulls images + audio (~15 MB)
Peak bandwidth	104.1 Mbps	~10% of the 1 Gbps line
Total data transferred	3.38 GB	over the run

3. Interpretation

600 users is well within capacity. The server stayed at 99.94% success with warm responses under 263 ms and used only ~10% of the

network line. There is significant headroom above 600.

- **Warm experience is fast:** once a user's first page is loaded, navigation is ~263 ms at p95.
- **First-visit (cold) load is the heaviest moment:** p95 ~20s because the first page pulls images and audio (~15 MB). This is a one-time cost per user, eased by the asset diet already shipped and by staggering entry.
- **Network is not the bottleneck:** peak ~104.1 Mbps is a small fraction of 1 Gbps.
- **AI rotation safe:** the live 10-key AI Studio pool covers generation load separately; this test confirms the web/serving tier.

4. Recommendations for Thursday (600 people)

- **Stagger entry** over 1-2 minutes if possible, to smooth the simultaneous cold-load spike.
- **Stand by** at class start: monitor container health and error logs live.
- **No action needed on capacity** — 600 is comfortably handled.

Generated 2026-06-08 · 600-user load test · Data: loadtest/summary-600-fresh-2026-06-08.json (k6) · Test ran from a single client machine; server-side bandwidth headroom is larger than the figure shown.